

Consumed with Greed

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Repository Link: https://github.com/jmarison/COMP323_GroupProject.git

Build/run target:

- Tested on Mac and Windows 11
- Compatible with Pygame 2.6
- Compatible with Python 3.x

Demo Goal:

The goal for the demo is to showcase the core gameplay loop, a fast-paced roguelite dungeon crawler. The demo aims to show that the procedural dungeon generation, combat system, and player progression mechanics (Items/Weapons) work together to create a fun, playable, and engaging experience. Within 2–3 minutes of play, the audience should understand how the player moves through rooms, fights enemies, collects coins, manages weapons, and most importantly how divine protection functions. For this demo, the scope includes procedural dungeon rooms, enemy AI, weapons, combat, coins, and the HUD. Systems such as full progression through multiple floors and the floor boss are outside the scope of today's demo.

Controls and Player task:

Controls: (Controls are rebindable in main menu and saved to settings.json for future play)

Move: WASD

Aim: IJKL

Switch Weapons: E and Q

Restart Run: Y

Debug Toggle: F1

Player Task:

The player is placed within a generated dungeon with a pistol and dagger. They are tasked with collecting coins, weapons, and items to become increasingly strong to fend off enemies and the floor boss. The player's damage multiplier is tied to their coin count, forcing them to choose between spending coins on upgrades or saving them to maintain higher damage. The player loses upon hitting zero HP. The game is designed to be endless so there is no definitive "win" state but the floor is considered "won" once the floor boss is defeated.

Playable Demo Script:

* all sections presented by Jacob Marison *

Introduction: (0:00 - 0:30)

Consumed with Greed is a top-down dungeon crawler where the player is tasked with collecting money, items, and weapons to become increasingly more powerful to defeat enemies and bosses.

Dungeon Generation: (in spawn room explain) (0:30 - 1:45)

Dungeons are created using A* algorithm. Each Dungeon has 1 spawn room, 1 boss room, 1 mini game room, 1 weapon shop room, and 10 normal rooms. Normal rooms are a typical room of enemies from a handful of layouts. Upon entering, the doors lock forcing the player to clear the room before progressing. Mini game rooms are where the player gets to choose between 3 items to purchase. Weapon shops are the same as mini game rooms but with weapons. And the boss room currently does not have a boss, but when it does it will be required to beat and afterwards will give a door to the next dungeon.

Combat: (enter an uncleared room) (1:45 - 2:45)

The player begins each run with a pistol and a dagger. The pistol and other ranged weapons shoot out a bullet which can pierce enemies depending on the weapon. Ranged weapons have a limited ammo supply which cannot be restored. Melee weapons attack in an arc in front of the player with infinite ammo.

Divine Protection and Money: (second uncleared room) (2:45 - 4:15)

Upon entering a room the player receives “divine protection.” This blocks a single instance of damage for the player. Typically when an enemy is defeated they have a 33% chance to drop a coin for the player, but if the player still has their “divine protection” (meaning they defeated the enemy without being hit) then each enemy will have a 100% chance to drop coins.

The amount of money the player has is directly linked to their damage multiplier. So as the player collects more money they become increasingly stronger. As a result they must choose whether to spend their money on weapons and items for flat stats (damage, bullet speed, player size etc) at the expense of their damage multiplier or keep their raw damage. This rewards players who can maintain the divine protection and collect more coins while still allowing those who lose it to still collect some coins.

Weapon and Item rooms: (Whenever enter weapon/item room) (4:15 - 5:00)

Weapon and Item rooms have 3 pedestals with a selection of either 3 items or weapons (depending on room type) for the player to choose from. The player can choose one and the others will disappear. All weapons and items sell for 10 coins each. The player has a max weapon inventory of 2 weapons, when buying a weapon at max capacity it replaces the weapon in hand.

Boss Room and Floor progression: (Whenever enter boss room) (5:00 - 5:30)

At the end of the dungeon is the boss room. The boss is currently a WIP, but the room still has the 'goal' which sends the player to the next dungeon, continuing the loop. A new dungeon is generated and the player keeps their collected items, weapons, and coins.

Future Plans: (5:30 - 6:15)

Boss fights with 2 bosses to pick from; more weapons, items, enemy types, and room layouts; Visuals and SFX for weapons/enemies; difficulty scaling across floors; flesh out player (currently thinking of a money bag which goes around literally consuming money).

Architecture map (scenes and modules):

main.py

Initializes pygame

Starts game loop

game.py

Main update and draw logic

Handles player state, run resets, and floor progression

player.py

Player movement, combat, health, weapon inventory, item modifiers

entities.py

Enemies, bullets, hazards, coins

room.py

Room structure, walls, doors, enemy spawning

room_layouts.py

Holds all the hard-coded normal, mini game, and weapon shop room layouts

dungeon_generator.py

Procedural dungeon layout generation

item.py

Items catalogue and pedestal system

weapon.py

Weapons catalogue, mechanics and firing behavior

keybindings.py

Stores default controls and handles saving/changing controls

Input Handling

Player input handled inside player.update()

Update Loop

game.update() updates player, enemies, and dungeon state

Draw Loop

game.draw() renders rooms, entities, and HUD

Reliability Checklist:

- runs from a clean folder (no missing assets) – yes
- runs after a restart (quit → run again) – yes
- title → play → gameover → restart works – yes
- inputs are gated by scene/state (no movement on title screen) – yes
- FPS is stable on your machine (no gradual slowdown) – yes
- no object leaks (projectiles/enemies clean up or respawn) – yes
- README run steps are correct – yes
- controls are shown (title screen or README) – shown in settings menu

- window size works on the projector (text readable) – yes (same size as class exercises)

Playtest + Feedback plan:

Did any rooms feel unfair? (accidentally bought item on entry, unmissable enemy on entry)

Did the scaling feel natural? If struggling to maintain protection, did it feel impossible to collect the coins needed for damage, weapons, and items?

Clarity and HUD:

- Did the HUD ever feel like too much clutter? Does it cover enemies/player too much?
- Do immunity frames feel long enough? If you hit a hazard do you have enough time to escape it before becoming hittable again?
- Should items always display what they do (currently in debug only) or would it feel like too much clutter?

Control Feel:

- Did the player feel tedious maneuvering around hazards? Are door warp zones awkward to enter (ie too small)?
- Would you rather aim keys be default to IJKL or arrow keys?

Contributions:

Jacob - everything (demo walkthrough, project structure, all of code base, etc)